Active Inference Institute ~ 2024 Quarterly Roundtable #2

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All right, hello and welcome. It's June 28th, 2024. We're here in quarterly roundtable number two for this year. Welcome to the Active Inference Institute. We're a participatory online institute that is communicating, learning, and practicing applied active inference. Find us at some of the links here. This is recorded in an archived live stream. Please provide feedback so we can improve our work. All backgrounds and perspectives are welcome and we'll follow video etiquette for live streams. And also I'll show this new slide with some nice stable links that will be good signposts and stable ones too. So we're in our second quarterly roundtable of the year. There's going to be three main sections. First, institute updates. Then we'll go to the two organizational units and talk about project scale updates. And then look a little bit towards the second half of 2024. First though, anyone who wants to unmute and just go ahead and introduce. Say hi and anything else you want to begin with. I'll break the ice. Hi, I'm Blue. I am really excited at all of the things that the institute has accomplished this quarter. The fellows program, the partnerships, and our upcoming philanthropy program, TBD. More later. I'll go next. I'm John Boyk, one of the new fellows this year. So excited to be joining the institute in that capacity. And I guess there's more later. I'm Jean-Francois Cloutier. I'm also a research fellow, part of that first batch. And I'll echo John's words that I'm very excited to be in this role and hope that I will be able to provide some good value. Hi, I'm Andrew. I'm an intern with the institute. I've been involved with a few different projects on and off over the past maybe year, especially helping facilitate the textbook group whenever I can. Looking forward to that coming back up in a couple of weeks. And yeah, looking forward to seeing the rest of the developments with the institute, seeing it grow. Okay. Is that good next? Yeah, real quick. Yes, my name is Holly Grimm. And I'm a second member of the scientific advisory board for the second year. I'm working with the journal project with Alex. And right now, I'm kind of building a set of utilities in Python and Russ to help transcribe all of Daniel's amazing YouTube videos and sessions and trying to, so we can eventually create a way of using retrieval augmented generation, creating an LLM agent or chat with all of these sessions that have been recorded. So, thanks.

Alex Grimm. Okay, let me jump in. I'm Alex, one of the officers of the institute, also into journal project and some hours. So great to see everyone. Let's have a this round table.

Awesome. Okay. First, going over some institute scale updates on the digital infrastructure side, as the stable links are kind of the tip of the iceberg of, we'll be looking and have been making some updates to the accessibility and usability of the different services at the institute.

One of those stable links will take you to ecosystem.activeinference.institute. And at this page, which again, the endpoint may vary, we'll look to develop and really communicate more about what is the Active Inference Institute? What is the ecosystem? How are they articulated? How have they begun and changed through time structurally and all these different ways? And so one useful piece of information at that resource is kind of like a menu or kind of continents of the ecosystem. And another exciting announcement is that we'll be doing an updating process on our 2023 institute slash ecosystem paper. So it'll be very fun. There will be a lot of multimedia and collective exploration and refinements of big, impactful visions and things that people are motivated and excited and already working on. And so to be able to clarify that within the second half of the year in a certain way and also understand what what is the role of different functionalities in the ecosystem at the institute. So this will be a fun process in the second half of the year that we'll share more information about, but it'll involve all aspects of the institute. Anyone want to add anything here?

Okay. So thanks to the officers, the board of directors who had their second quarterly meeting several weeks ago, and to the scientific advisory board who completed their first phase of activities and we'll head into phase two soon. At the website, there's more information about these positions for now. Thanks to everybody who's been carrying it on this quarter. One of the really exciting new aspects and programs at the institute in the last few weeks was the introduction of the research fellows program.

So perhaps, JF and then John, what is it in how you're seeing it and what you'll like to do with it?

I'll go ahead. Well, I've been conducting a research project here at the institute for the last two years, now called the symbolic cognitive robotics project. And this project will now continue continue under the umbrella of a research fellowship. And what I hope to gain from this is increased participation, involvement, exposure, which will bring in unanticipated rewards. I'm sure it's already started. And what I hope to provide is to use Daniel's words is increased epistemic value of my research, and hope to be able to present on a more regular basis, my progress and, and hoping that the institute will benefit from providing me this, this, this increased exposure.

Daniel's words, again, my name is John Boyk, and my effort is focused on cognition in the large group setting. So how, how can a group of, say, 100,000 people or a million people come together in some kind of online ecosystem to make decisions, discuss events,

and make other cognitive and perform other cognitive processes? How does that happen efficiently? How can everyone have their rich input heard?

Daniel's words, I mean, I think, in the large group, and the group in my work, the group might be just a group of people focused on some issue, or it could be actually something more formal, a formal group like a city or even a nation, perhaps one day. So that's, that's what I'm working on. I hope that's, will be productive and useful.

Awesome. And Anna Pereira will share more about her work in the weeks to come.

This is an exciting program. It's, as with all things, on a rolling application basis. So reach out if you have any questions or to complete the application.

That's about individuals who are learning and applying active inference. Also on the inter-organizational scale, we have begun the partnerships program.

So these are structured relationships between the Institute and other organizations. And we expect and prefer this will look like many things.

Already in the one accepted application, which we'll announce in the coming weeks and several that are in progress.

It's really interesting and useful even already to explore where people are at and understand how to respectfully and appropriately have a partnership.

Anyone want to add any comments or thoughts on this?

All right. And one major piece of news from the last several weeks was receiving the 501c3 or USA tax exempt status.

So the link donate.active inference.institute is one place that will develop into an on-ramp for providing different kinds of financial support.

We're definitely looking to learn and to collaborate and to understand how we can support the sustainability of the Institute.

And through that support the accessibility and utility of active inference.

Anyone want to make any comments on this?

I just wanted to thank Freed Frank, who was our they were our pro bono attorney that really worked hard over a long time to help us achieve this status.

Thank you, Blue. Important to mention, too. Thank you a lot.

On different digital platforms, we continue being active and growing bit by bit.

As we turn now to the middle of this roundtable where we'll look at project scale updates and through systems engineering and Alex's in particular vision for understanding the

for understanding this hierarchical nested nature of our ecosystem organization and project-based

teams in 2024 following chris fields's information physics information processing uh course we

decided to jump in fully to the prepare measure paradigm so here's alice and bob or system a and b

communicating and this could be communication of a thing through time could be across a communications

interface and so we thought about different ways to use active inference to learn and apply active

inference and saw a connection here between the project-based teams and organizational approach

and to connect that to the quantum active inference prepare measure cycle and so it's only been half of

a year however it's been a very interesting dynamic already with quite quickly in the beginning of the

year and then also continuing people submitted project preparations and and updated whether it

was a institute project that was just kind of re-entering the system or projects from people we

knew projects from people we didn't know various people completed these project proposal forms and

their projects were listed as active and then just in the last several days with many people's measure

forms being submitted the rest of this round table and the newsletter which is basically the same

information broadly same updates is prepared essentially in a copy-paste format from what people have

reported at the project scale in their own learning so this is really exciting to see

not just the code of forms capacity the kind of digital affordances that can be implemented various ways

but really the fact that people want to and are submitting more and more parts of their experience

on projects of their own other projects and so on so i look forward to this continuing to increase

the use and across multiple scales new and creative uses of the preparation the measurements it's a really

just exciting way to kind of not just learn by doing but be doing by doing and that's pretty cool with

active inference any general thoughts on this or or anything quantum active

systems and now even quantum information processing and it's possibly could look simple

like to do new entities prepare measurement forms but i believe as they are based on really like real physics with the current state of the physics it will bring into organization into institute a great affordance to develop based on it

states states

affordances, Ana Magdalena Ortado's work on Unifisica, we'll get to these updates, Pablo FM and others with Numinia, and Shauna Dobson, Hector Manrique, and Dean, and others in the math art conversation. So those are projects that facilitators have just stepped up to do, and several of the core educational projects at the Institute, the Active Inference Ontology, Audio-Visual Production, Textbook Group, and The Journal. So let's hear a little bit more about those projects. On the Textbook Group, Cohort 6 will continue with the second half of the book, and Cohort 7 will begin from the beginning of the book on July 15th. It'll be a fun go-around again. It'll have some classic and familiar characters like the acting fontology and using different kinds of question-based methods for learning and asking fun collaborative discussions. There'll also be some different elements. It'll be on a three-week rhythm. We'll do two weeks per chapter, and then each of the cohorts will have a third week. In the second half of the book, that third week will be more about applying Active Inference, looking more at the code, which is what the second half of the book is about. And in the third week of the first half of the book, we'll explore more what that might look like. And also with collaboration with Andrew Pendlin, we will have some additional math learning collaborations. And there's a lot of fun, continued work on the textbook. So it'll be fun to have had that month or two off, jump back into second half of the year with the Textbook Group. Any other textbook group thoughts? All right. In production, we've had some fun videos over the last couple months. If anyone wants to call out any specific live stream that they participated in or viewed and thought was fun, go for it. There was nine guest streams. There was an art stream, six insights, excellently planned and executed by Darius, and including the Ian McGilchrist surprise, which is, I think, happening concurrently.

There were several math art streams that were very creative and very exciting. We did a paper-based live stream series, number 57, which was the first in a few months on the paper by Parr, Friston, and Zeidman, Active Data Selection. That was extremely relevant and a really promising direction for the development of active inference. And there was an art stream with Nathan Schneider on governable spaces. That was also really cool. So anyone want to add any other thoughts?

Okay. On other updates from EduActive, from the Fundamentals of Active Inference project, we concluded a phase of work with Sanjeev in his incredible odyssey of the Fundamentals of Active Inference.

We concluded essentially weekly meetings since the beginning of 2023, and we look forward to next phases for that project and work. Ana Magdalena Ortada reports from Unaphysica project, and I'm sure we'll hear more from their work. Susan Hastie's weekly meetings also came to an excellent closure in the Action Research on Collective Foraging project as she heads into some new programs and directions for the work.

Pablo FM reported from the Numinia project. From one of the symposia, there was a demonstration of an active inference game in an escape room. In the same spirit has been built, which is kind of fun, and people can check it out in the newsletter where everything is for all the links and all that.

From the Math Art project, so kind of in the same forest as the live streams from Math Art, but part of the broader conversation. Bert, who continues to do awesome work in the internship, made a short report that I know there's so much more to say on. And from the Journal project, Alex and Holly want to share anything or what is exciting from Journal?

Yeah, so I kind of mentioned it already that I'm interested in doing retrieval augmented generation, and so I was able to get a huge one terabyte disk from you, Daniel, and take all those videos and transcribe them using a local whisper model on my server here at home. So that was nice in terms of not having to pay too much money out to get those transcripts done by third-party tooling. And yeah, I'm using open source models so we can start to, you know, make use of those transcripts. And I'm pretty excited about that.

So Alex, anything you want to add?

No, great. Yes.

Thank you. All right. To re-inference. All right. Just kind of snapshotting what projects we have that are active. So essentially those that have provided us a measurement and any that aren't here can be brought back on by measurement. From the ecosystem, Jeremy Cooper and others have been working on active inference account of belief updating and PTSD. Michael Carle and others have been working on the active inference model of translations for language. Jeff mentioned the symbolic cognitive robotics and Shiger Raman has continued work in writing on the humanity story of uncertain self.

At the Institute, we continue faster and slower and differently on knowledge engineering and seeing different opportunities for the kinds of knowledge engineering approaches that might benefit active inference in the ecosystem. Active Blockference, combining active inference with the CAD CAD systems engineering framework by BlockScience is continuing to proceed on a few different directions. And then one of our major activities in re-inference has been the RxInfer Learning and Development group.

So just to sort of contextualize this, in Active Model Stream 4.1, Implementing Active Inference by Message Passing in a Factor Graph from August 2021, I looked back to see this presentation by BiasLab and see what they

had planned to have done for the future of applying reactive message passing. Which at that point

was a research advance on top of the factor graph formalism was a research advance on top of the factor graph formalism. The Forney factor graph,

constrained Forney factor graph and so on, had enabled better logistics to go from the Bayesian graph to the variational message passing.

And then this reactive feature was a research advance on top of the

And then this reactive feature was getting added in. And so it's just really interesting to see how they laid out ForneyLab 2.0, ForneyLab.jl 2.0.

And then come 2024, we're having great times in the engineering channel in Discord and on a lot of people's personal projects.

So, Cobus's work at learnable loop. People interested in using Rx and for different kinds of hackathons and different projects.

And then also a crossover moment with the textbook group was with the help of some members of the BiasLab.

Here's one of the early figures from the 2022 textbook of the frog jumping out of the hand or the something jumping out of the hand or not.

And then here's what the app model block for Rx and for looks like. And so seeing this kind of an isomorphic or mapping between the visual representation, the code, the analytical formalisms, the art, all these different edges.

That's what kind of the fun has always been suggested to be. So to start to see the operational improvements in reactive message passing and the way that the groups and the developers involved in this project are closing the gap between what is suggested in principle and in computer science proof of example,

and taking it to just an amazing engineering informed level. Anyone want to add anything on orc.

I want to say that it's really great development and it's good that we organized that group of people who are beginners on learning that.

We are learning that toolbox. And I believe that toolbox which should provide the importance to build agents and collective agents will be a base for some breakthrough in artificial intelligence and possibly we are putting some cornerstone for it.

Thanks Alex.

Thanks Alex. And thanks to the many participants who are just doing awesome discussion in the engineering channel contributions to the coda and to the GitHub directly.

Alright, from the Active Inference Ontology project, there have been quite a few updates. Dave Douglas, as always, leading the V of birds, has produced two amazing open source works that are almost like snapshots of really broad ontological inquiries.

So first, a single author paper aligning spatial web terms to SUMO, which is in reference to the IEEE P2874 standards draft and working group. Connecting that to SUMO, which has always been this kind of vision of incorporating multiple ontologies, including formal ones like SUMO.

And then also in a snapshot of our primary active inference ontology. We had a number of authors join this open source snapshot about aligning the active inference ontology to SUMO.

So starting to triangulate and constellate multiple ontologies along the way, learning a lot about using, communicating, languaging around them, and different functionalities and niches for different structures and visions of ontologies.

So that kind of like meta-onological active inference component.

And also bringing the active inference ontologies in a connected way to formality while also exploring in the very same project and in kind of other meetings and spaces, more of the informalizable, ineffable components.

So that's pretty cool to see.

Any ontology ideas?

Just a short, short note that it's okay to have like multiple, multiple scales, multiple scales of models and have meta, meta, meta models.

Yeah, let's see

updating belief dynamics. It's a computational psychiatry project focusing on PTSD. And something I really appreciate about the Institute is its way to kind of remotely bring together a variety of people and researchers of different backgrounds. That group includes a computational neuroscientist, a clinical therapist, and then you have someone like me. And we're all learning kind of as we're doing in order to try and recreate something like the belief dynamics of an individual has PTSD in the direction of developing treatments. So it's been a super valuable experience as well as on and off working with the Rx and FUR group as I continue to develop my own ability to kind of code active inference agents and simulations. So yeah, here, this project, active agents and active inference approach to agent-based modeling in the social sciences. I'll be giving that this year at the 10th International Conference on Computational Social Science being held at University of Pennsylvania. Basically, the tutorial attempts to address growing interest in more cognitively complex agents to be used in agent-based modeling as opposed to just strict if-then sort of agents. If this happens, then they necessarily do this and so on, giving them the ability to make inferences. So I've taken a lot of inspiration from work like your own work, Daniel, on the inference project. That was my first foray into like thinking, okay, I can model an individual agent, but what, how do you model multiple as they interact in an environment that changes, which then impacts them, getting a better grasp of the grasp of differentiation between internal and external states. It's been great, as well as another paper by some folks from the Scientific Advisory Board for the Institute, Mo Alvaricine and Connor Hines, which is also

really great, very social science oriented work. So I'm doing something in turn is kind of they did, I created a simulation that recreates a common social science paradigm, multiple agents working in a network, engaging in parallel problem solving, and it using active inference, it recreates the dynamics of the original paradigm, if you were to run it through that logo or some other common software.

but uh thanks for including this in the uh in the talk here so awesome cool and helping facilitate

the previous interval of the textbook group too
and and with ali on that too jeff sure uh so the uh symbolic cognitive robotics project
um asks the question um what does it take at a minimum for a a simple mobile robot to survive
in a world it knows initially very little about so the focus is is in on learning through engagement
with an agent's environment it it focuses on the development of agency uh in a robot which starts
with very little in terms of uh cognitive constructs and through engagement through acquisition of
experience grows a um a society of mind a a collective of simple cognition actors uh
which each one uh quite uh specialized in terms of what it can perceive and what it can do
um and and they all work together in the sense that a cognition actor will through observations of its
umwelt uh generate synthesized beliefs which will then become observations to more abstract cognition
actors so you have this ladder of belief abstraction uh that is built up through active engagement in the
environment through uh learning what simple perceptions mean learning what affordances there are
what can the robot do um and as the robot grows its collective of cognition actors uh the conjecture is that
it will detect uh trends and uh how its its its beliefs its more and more abstract beliefs map uh homeostatic signals
and uh how its its beliefs are um am i doing something that benefits my survival in my continuance
continued survival so it's really a project about um uh how an agent learns its own agency learns its
affordances learns to make sense of its perceptions through more and more elaborate beliefs
beliefs um and and all of this sense making is grounded in the survival imperative now of course the the
physical robot is not mortal i mean i can turn it off turn it back on it's still there it's like your
computer but this societal minded will grow through its its its experiences through its learning is itself
mortal and i set up a a an experimental environment such that the robot as such as as a minimum
amount of time in order to grow competencies at exploring and foraging otherwise it expands its its store of
energy without ever replenishing it so it needs to forge to replenish it needs also to avoid collisions because
collisions are deleterious to its health it it needs also to keep a level of engagement otherwise it it it
essentially retreats away from the environment so the all these motivations um uh activate
this this this this this um society of mind that the robot uh grows over time and all i'm trying to do is
set up this mechanisms these mechanisms and launch the robot and see if it can grow it's it's it's it's make
up its own mind basically about the world it needs to engage uh with so there are notions of collective
intelligence there are notions of mortal computing there are notions of uh grounded sense making uh affordance
discovery um there's also um a very big nod uh to uh emmanuel kant and it's in his epistemology because the
the the robot is is needs to develop causal theories of its perceptions how do i make how can i
predict what i predict what i will perceive next is is very important so a very strong notion of
predictive processing based on the discovery of causal theory how do i make sense uh of of the
perceptions and the actions that i that i that i engage into how can it how can i predict what happens next
and all these concepts are are brought together in in in a um in a hole that hopefully uh leads to uh
autonomous agency and really simple lego robots which uh brings in a dimension of fun as well because

these
robots are are really fun to build and fun to run um and my hope is that not only does uh the robot uh get
to learn how to survive but that we can observe its behaviors and its observe and its visible states and
subject them to an active inference analysis and be able to demonstrate that yes indeed the robot uh reduces
its varicul free energy and gets better at it over time or not so it's it's a very fun project and i um i i i
get a lot of uh uh input from the community and uh i'm constantly fed by all the material
that the institute uh produces so it's um it's a very fun thing to do awesome thank you both for the
updates all right that'll be uh oh yeah sorry yeah go ahead please please yeah as for andrew's work uh i
believe at some point it could be integrated into social sciences sciences course somehow if it's
interesting possibly it's uh could be not not only another chapter in the course but also somehow
augment our chapters with some computational approach and eric's infer practice and i want to say big
thanks for jayef for his continuous work on his robotics projects it's really great and really great to see
in its development through the time thank you very much thank you i agree with that anyone else want to
make any comments on reinference or projects i'd ask a question to jf if i if we have a second can you just
uh talk a little bit about your use of the term symbolic uh in symbolic cognitive yes so the um
the causal theories that uh each that the the robot discovers and i i have to say uh
its mind is a society of cognition actors and each cognition actor will want to develop a causal theory to
predict predict what it will observe next in its own limited umwelt those causal theories are essentially
logic programs that it discovers uh i will observe c if i've i'm observing a and b for example um
so uh much of the learning done by the robot are are captured as either um
um logic programs which are uh symbolic or and or as uh beliefs which are also uh uh symbolic predicates
so uh the the the the world view that the robot develops internally intrinsically is expressed
symbolically as these logic programs and and these predicates
the symbolic uh in what language uh the uh i use two programming languages for the the symbolic processing uh
and discovery of of of of of uh causal theories i use prolog which is very well suited uh to uh state uh
uh to to express uh state search uh search uh problems uh search space uh problems and to solving them
and i use a functional pro language called elixir to implement uh perception and and action basically
interfacing with the robots percept uh sensors and uh actuators prologue and elixir thank you
all right as we head into the last part of the stream so if anyone has questions from the live chat they
can write it otherwise i will make some notes on one slide and then anyone who wants to bring up anything
or ask anything or or anything that they're excited about for the second half of the year or however they
can go for it so um in the hour before this stream ends ends on the verge of a loss of nothing to share
i just brought in some pictures from the prior weeks and wanted to open this space and have the time of
reflection and this multi-scale self and multi-scale time and space concepts and just where have we been
we're in the round table where are we at where are we heading and then where when how etc is active
inference on that and and just a few of the adventures of the last week well first where where were we in
the last end of the last round table and here we had these multiple horizons and what was happening in

active inference and what we could be doing and a a visit to host archival materials and a visit to view
bucky fuller's archive materials both really impressed upon me the the relevance and necessity of
documentation archiving intergenerational cultural contact and many more things and this was a funny
picture taken from a diner we deliver norms but we don't only deliver norms we're also part of them
and so on so that was kind of a funny sign to see and then just in terms of
how to move into the next half of the year
so that's all i'll say there how about just anything people want to ask or mention for coming forward
i have a question why is your bike called red queen oh just kind of moving faster and faster and staying
in the same place
anyone wanna
uh add anything or what yeah alex yeah i want to say say thanks to all everyone who participated today
and that's great that we bring that back to the format of uh participating more broad broad participants
than just of service and it will be great to see people next rounds so let's continue
uh
actually i had a quick question i used to be much more involved with the
engineering group the rx infer group and i fortunately had to sideline my participation with it
and there been any particularly interesting developments recently aside from the lights live stream
uh
what to say well other than look at the documents and their repos um
several things have been learned and developed kobus's work at learnable loop and also the developer
teams examples have very rapidly caught up with the major release 3.0 which was a somewhat breaking change
in the sense that the prior app models were changed and didn't work in 3.0 but
it's built in a better way so to see how fast the developers were able to iterate on the versioning
and then the learning of the model building and then to as we start to have people with more computer science
less more active inference less and engineering experience and all this and and all these
different perspectives and that helps to disentangle where are the paths that lead into
the analytical details of message passing how is that being implemented from what space of ways it could be
implemented so like what's required by rx infer and rx environments what's just a norm of use
emerging norm of use but not something required so a lot of this is still being fleshed out but
to see some new types of models open source and then just personally like to be able to take
kobus's examples copy them into an online cloud browser notebook hit run all reload the page hit
run all again and have a real drone active inference simulation schematic but educational
run in two clicks it's just something we didn't have one two three etc years ago so to be and for
that to be not just a wrapper or like a buffer around a simplified way but that rather to be
an interpretable engineering grade spec that's just brilliantly abstracted from the reactive message
passing it's a major tech unlock super cool um also bonus points that uh you can so

quickly reproduce the code uh in your own system or in your browser and run it that's what i'm doing

for my conference i'm doing everything in google collabs so that attendees can just hop in if you

have a google account you can do whatever you want uh i think that's a great sort of learning resource

for for institute members and others too just keep it keep it ready to go no need to dig through a ton of

uh you know library archives and subdirectories to finally get the thing running

yeah anyone else want to add or ask anything

yes since it's my first presence at this round table i'd just like to uh express that um

um how much i appreciate the openness and and and uh welcoming nature and the welcoming nature of

this community it really encourages multiple backgrounds multiple perspectives it listens uh and

and and it speaks in very in many voices and i find that it it really uh uh promotes

true open science and it's it's i don't take it for granted it is rare and i appreciate it

thank you jf

okay last call okay thank you everyone thanks everybody joining and looking forward to see you around so bye

thank you